CHALLENGES FACED AND SOLUTIONS IMPLEMENTED

Meka Hrishi Teja Chowdary

Introduction

During the development of the Raritone Backend System, several technical, organizational, and project management challenges were encountered. These challenges provided valuable learning opportunities and helped improve problem-solving, collaboration, planning, and execution skills.

1. Understanding a Large Project Structure

Challenge
Initially, understanding the overall project architecture and how different modules interacted with each other was challenging due to the project's size and multiple contributors.

Solution Implemented
The team spent time analyzing the project structure, understanding module responsibilities, and documenting workflows. This helped create a clear understanding of the system before integration.

2. Coordinating Work Among Multiple Team Members

Challenge
Different team members worked on different modules simultaneously, making coordination and synchronization difficult.

Solution Implemented
Regular communication was maintained to discuss progress, dependencies, and integration requirements. Responsibilities were clearly divided among team members.

3. Maintaining Consistency Across Modules

Challenge
Modules developed by different developers could have inconsistent coding styles, naming conventions, and API structures.

Solution Implemented
Common development practices and REST API conventions were followed to maintain consistency throughout the backend system.

4. Managing Frequent Requirement Changes

Challenge
As the project evolved, new modules such as Avatar and Product Mapping were introduced, requiring modifications to existing designs.

Solution Implemented
The backend was designed with modularity and flexibility in mind, allowing new features to be integrated without affecting existing functionality.

5. Learning New Technologies Quickly

Challenge
Some technologies and concepts used in the project were new and required additional learning and research.

Solution Implemented
Official documentation, technical articles, and practical experimentation were used to gain understanding and successfully implement required functionality.

6. Researching Future Technologies

Challenge
Technologies such as Redis, Docker, Kubernetes, Load Balancing, and AI integration were not directly implemented but required understanding for future planning.

Solution Implemented
Dedicated research was conducted to study industry-standard practices and evaluate how these technologies could benefit the platform in future versions

7. Preparing Multiple Deliverables

Challenge
The internship required not only development work but also documentation, reports, presentations, GitHub management, and research submissions.

Solution Implemented
Tasks were scheduled and prioritized to ensure that both development and documentation activities progressed simultaneously.

8. Understanding Real-World Software Development Practices

Challenge
Moving from academic projects to a larger collaborative project introduced unfamiliar development workflows.

Solution Implemented
Version control, branch management, module ownership, testing procedures, and documentation practices were followed to gain practical industry experience.

9. Managing Time Effectively

Challenge
Balancing development tasks, academic responsibilities, and internship deliverables required effective time management.

Solution Implemented
A structured approach was adopted to allocate dedicated time for coding, testing, research, and documentation activities.

10. Presenting Technical Work Clearly

Challenge
Converting technical implementations into professional reports, presentations, and documentation was challenging.

Solution Implemented
Complex technical concepts were simplified using structured diagrams, workflows, tables, and clear explanations for better understanding.

11. Planning for Future Expansion

Challenge
The system needed to support future features such as AI-generated avatars, recommendation engines, and advanced virtual try-on capabilities.

Solution Implemented
Future-ready schemas, workflows, and architecture designs were prepared to support seamless expansion in upcoming versions.

12. End-to-End Project Understanding

Challenge
Although individual team members worked on specific modules, understanding the complete project from start to finish was necessary for final documentation and presentation.

Solution Implemented
The complete backend system was reviewed, module interactions were analyzed, and comprehensive documentation was prepared covering the entire project.

Conclusion

The challenges encountered during the development of the Raritone Backend System extended beyond coding and included collaboration, planning, research, documentation, and system design. Successfully overcoming these challenges strengthened technical knowledge, communication skills, teamwork, project management abilities, and understanding of real-world software development practices.